

ITALIAN FINANCIAL CHALLENGE

AI-Powered Financial Health Classification

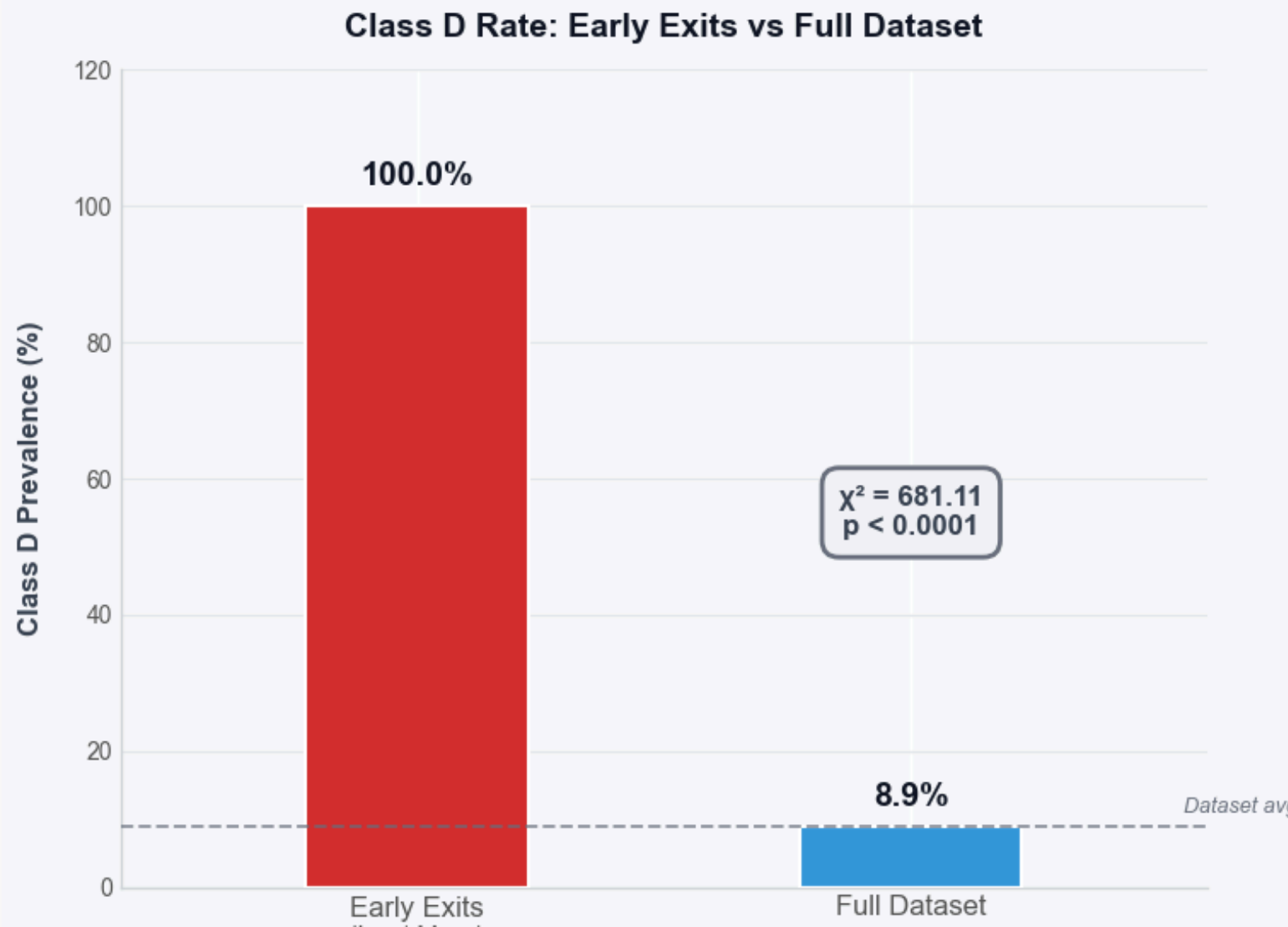
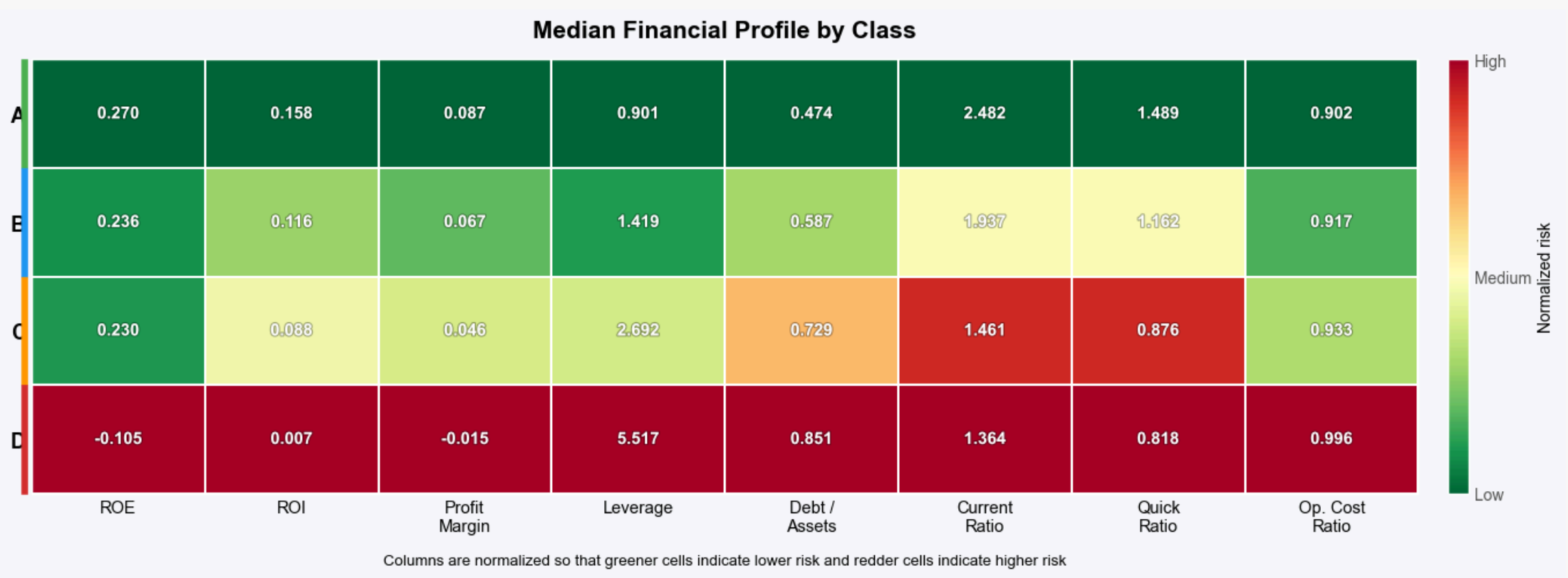
LUISS Guido Carli - Grand Challenge

April 27, 2026

Prof.ssa Finocchi Irene and Prof. Poccianti Cosimo

Direnzo Carla Moriondo Tommaso Pazienza Lorenzo Raimondo Eleonora

FROM THE EDA FINDINGS



Financial ratios provide clear class separation

Null ROE is an exclusive signal of Class D

Class distribution is stable over time

AGENDA

1. **Class separability**

Decision tree and fallback

2. **Spectral signature and geometry**

ArrowSpace spectral geometry

3. **Decision System**

Routing logic across economic regimes

4. **Temporal Validation**

Testing model stability over time (2018–2023)

5. **Results**

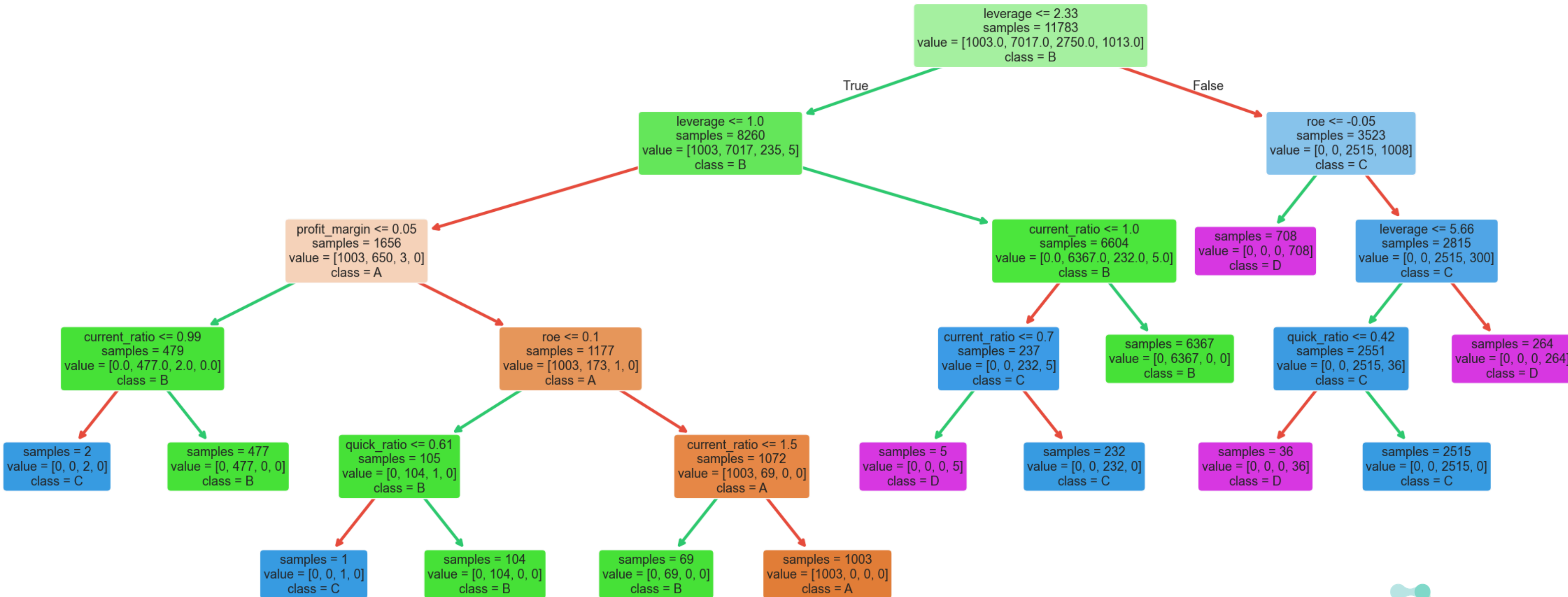
Test predictions and spectral anomalies

6. **Impact**

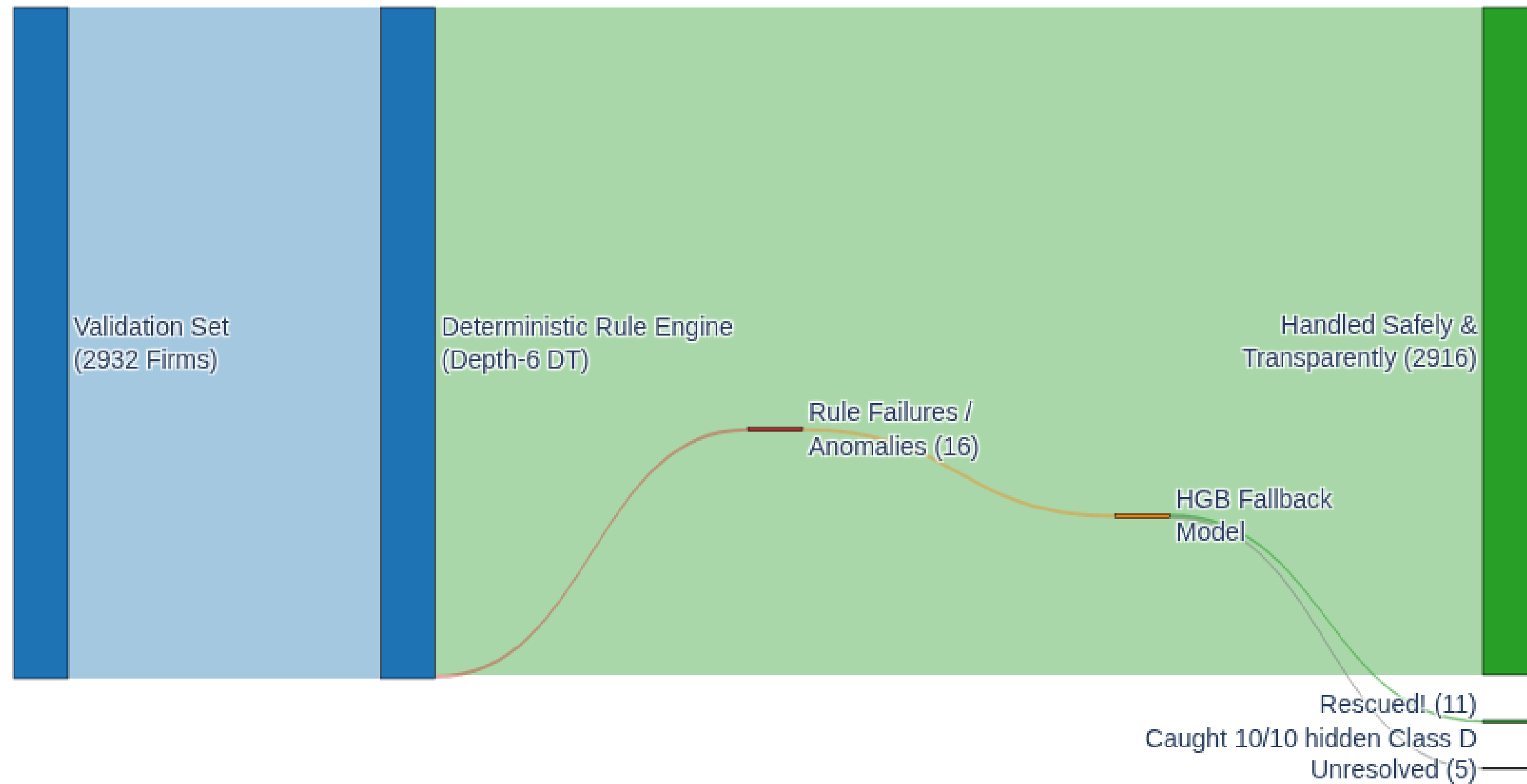
From model to deployable decision tool

DECISION TREE RULES

Decision Tree (max_depth=6): recovered labelling function



HGB FALL BACK



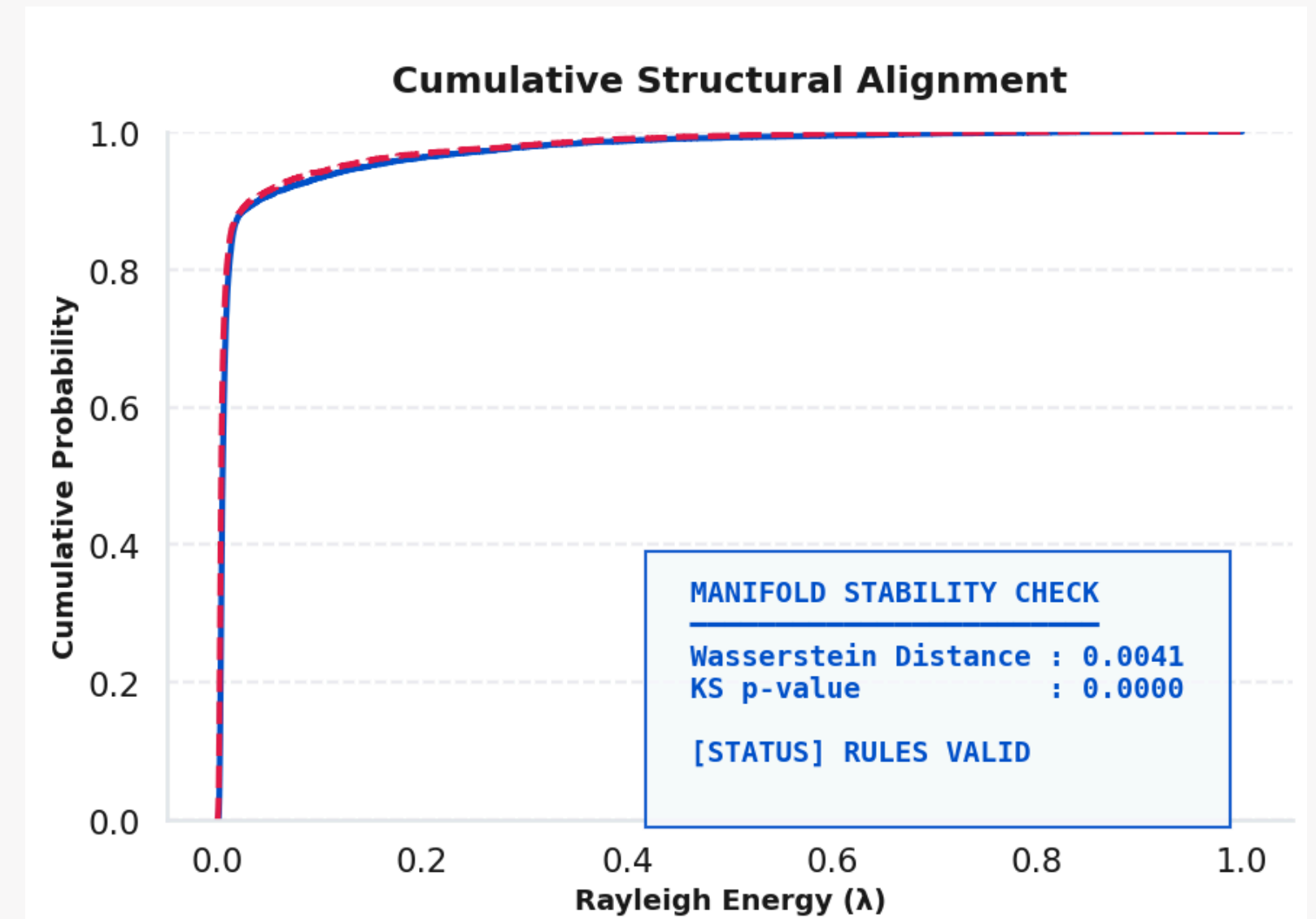
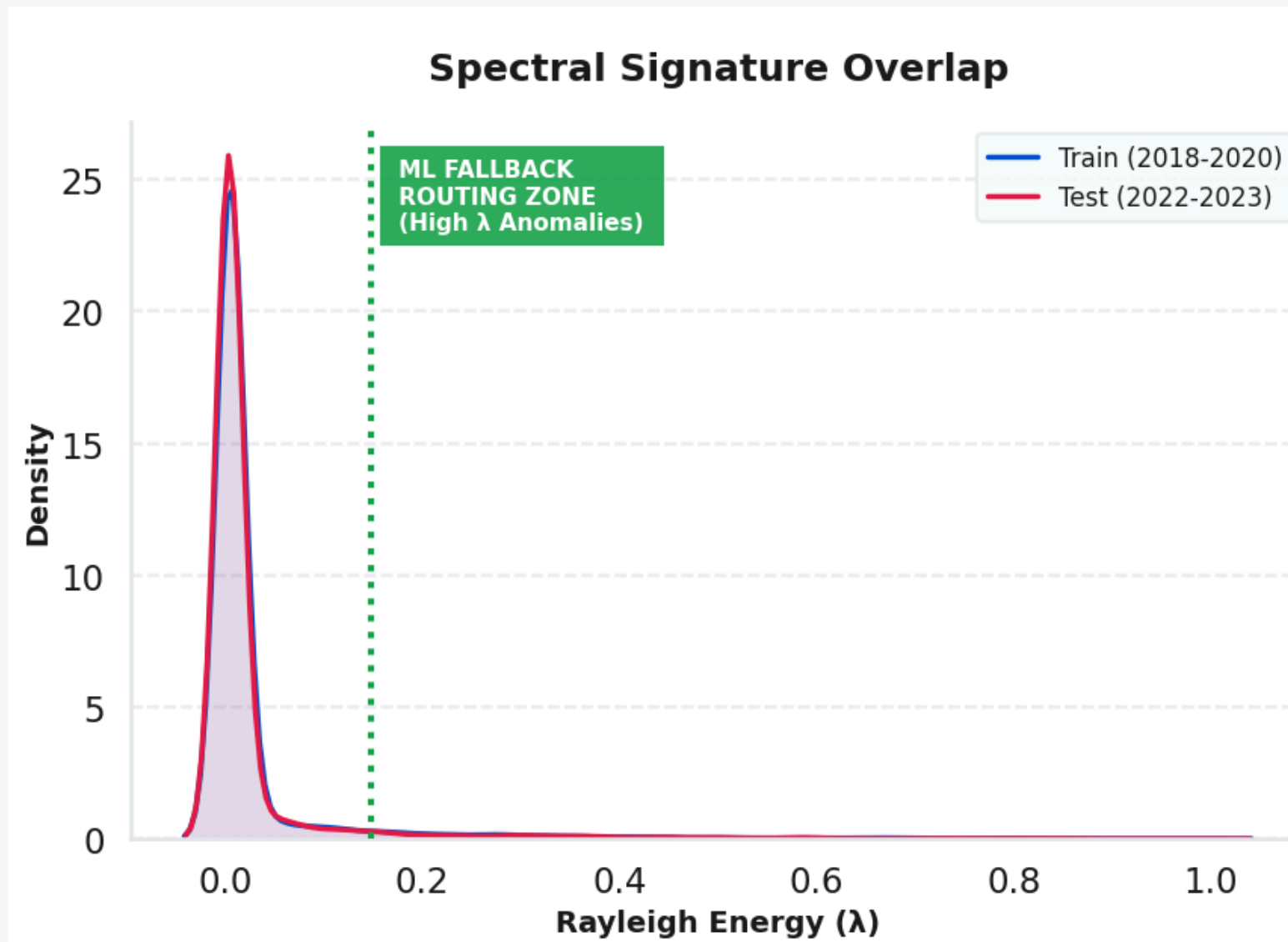
99.4%

handled by auditable rules

100%

hidden distress cases caught

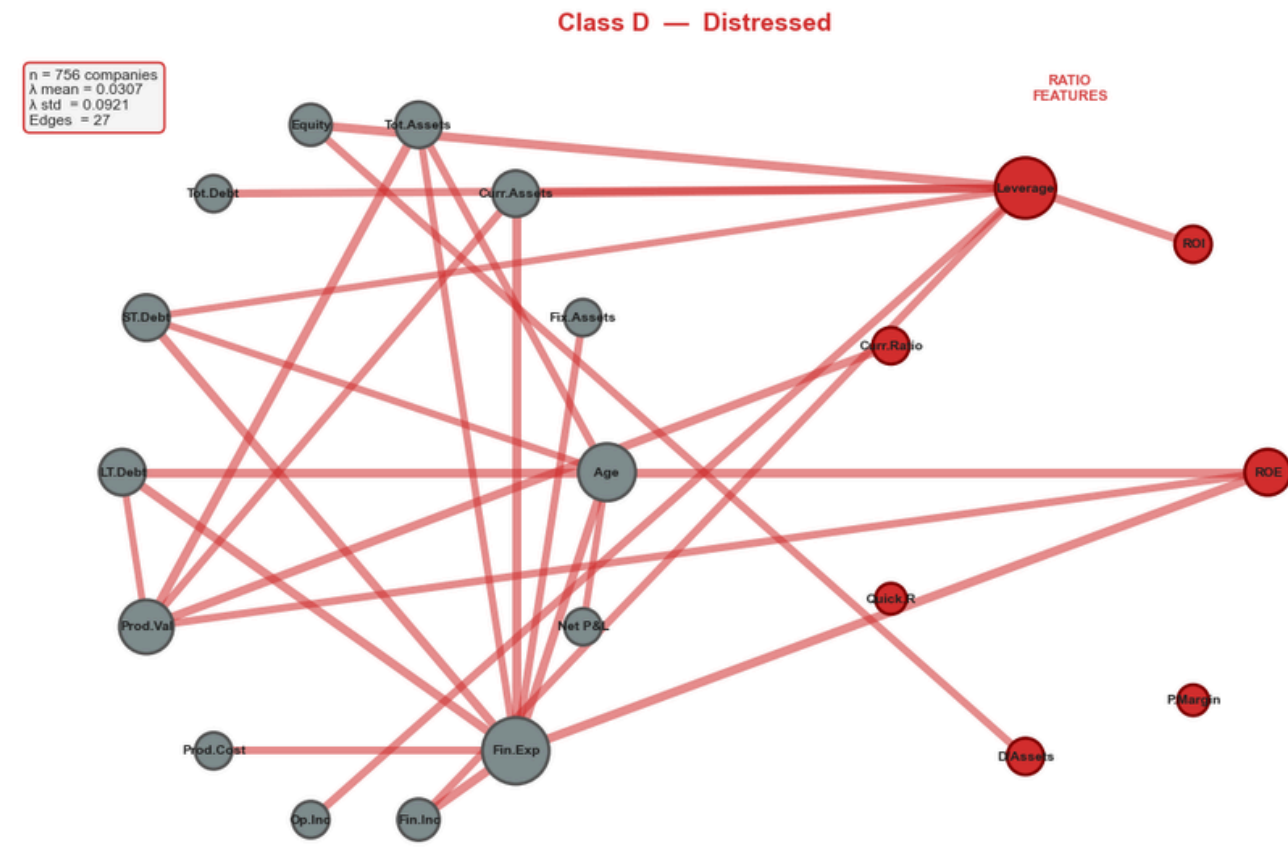
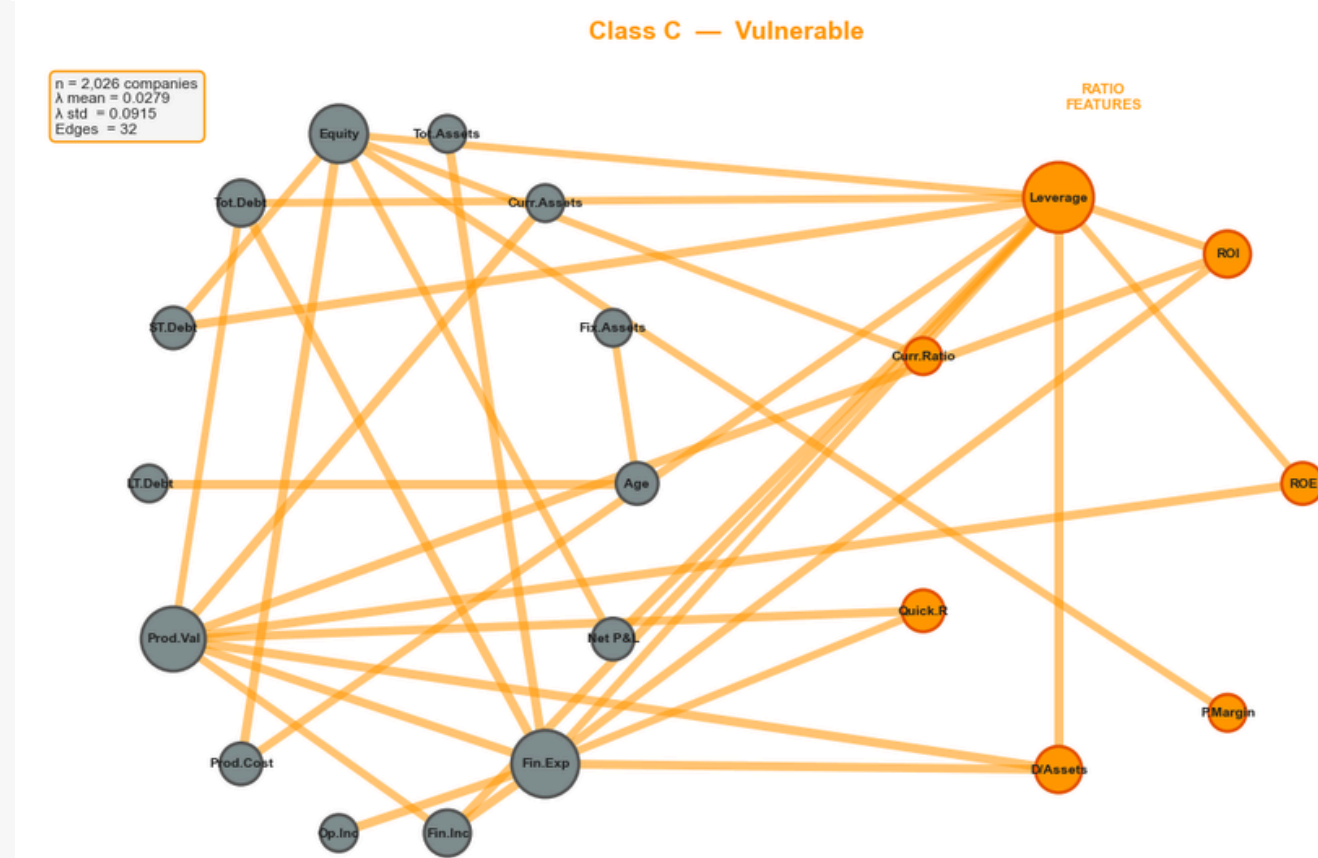
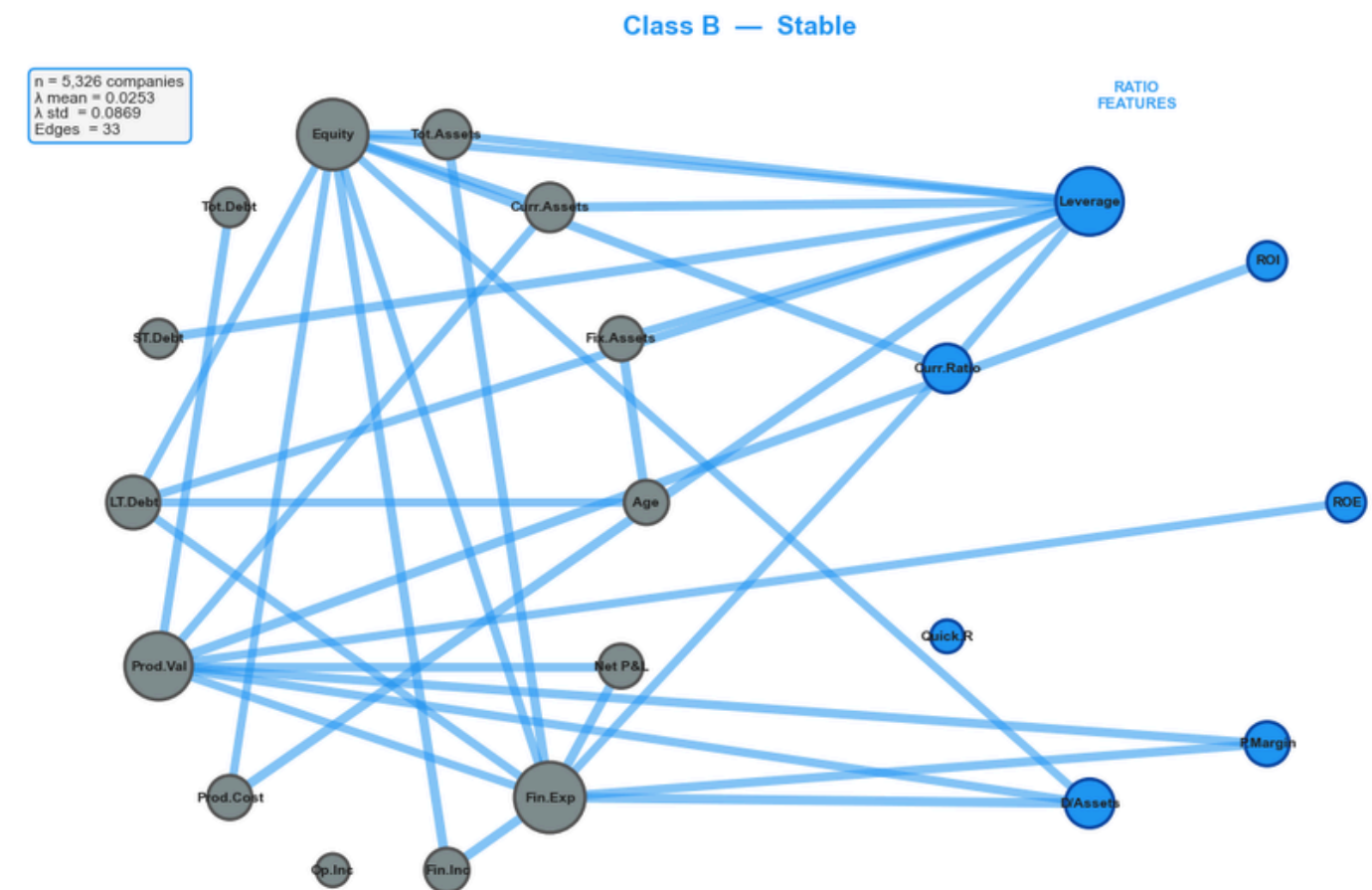
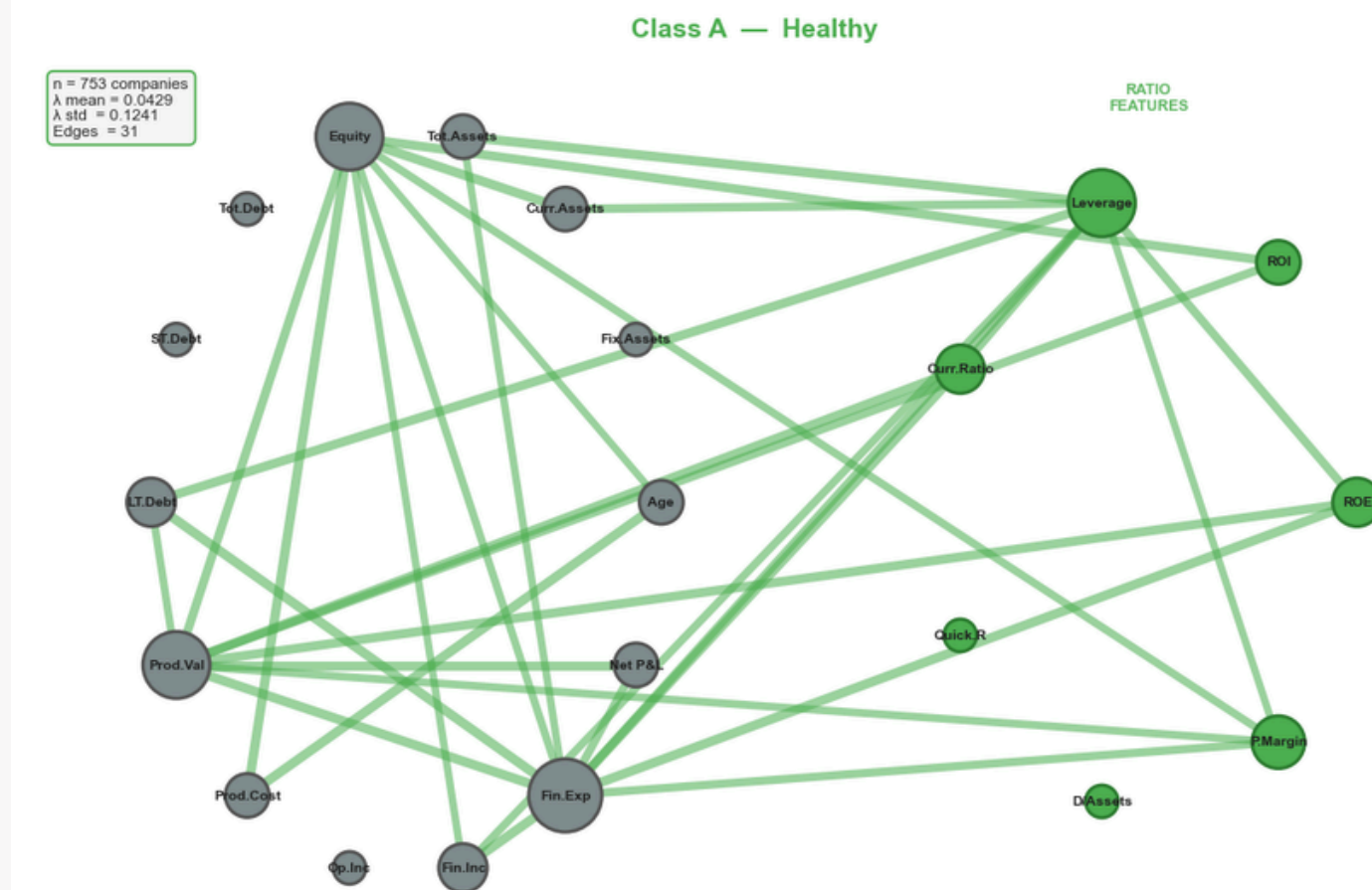
SPECTRAL SIGNATURE



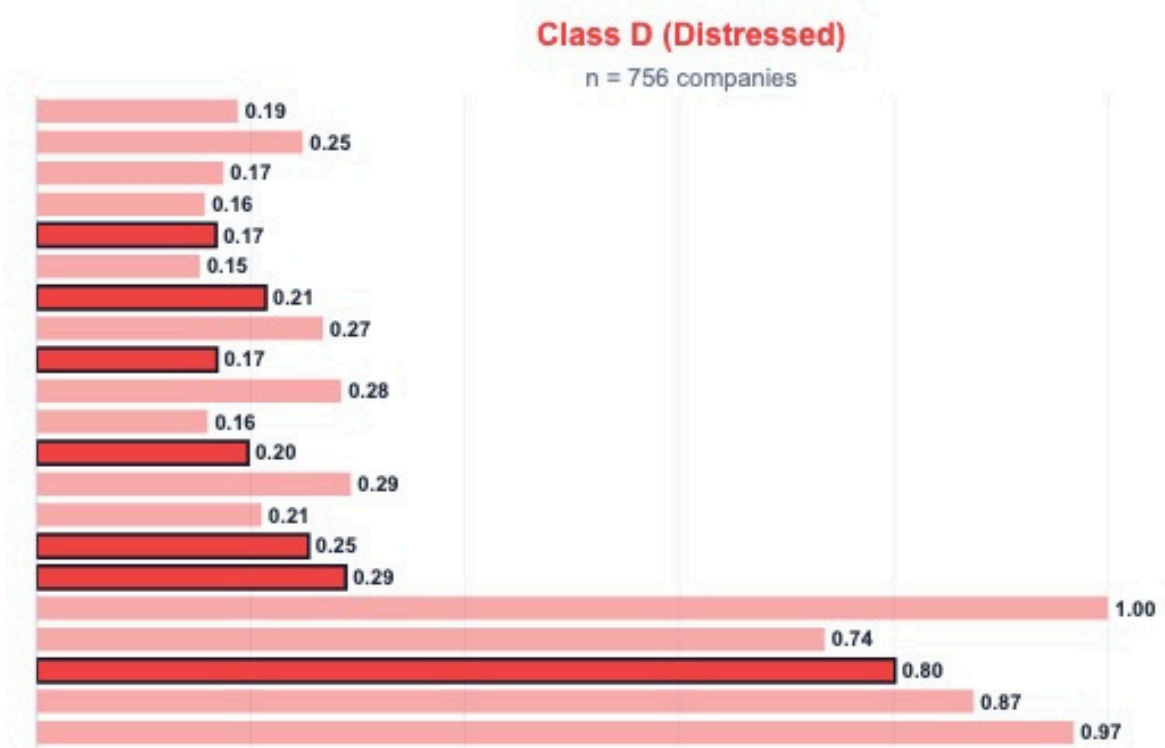
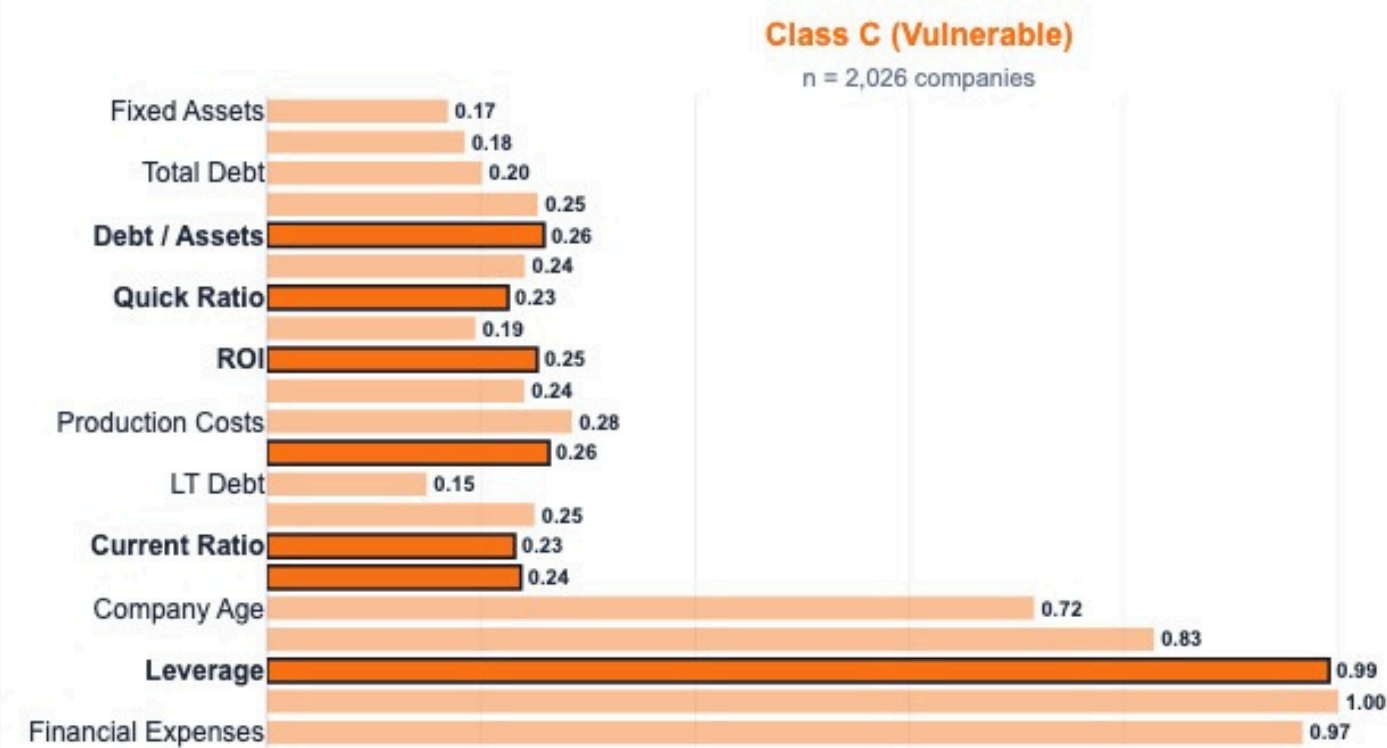
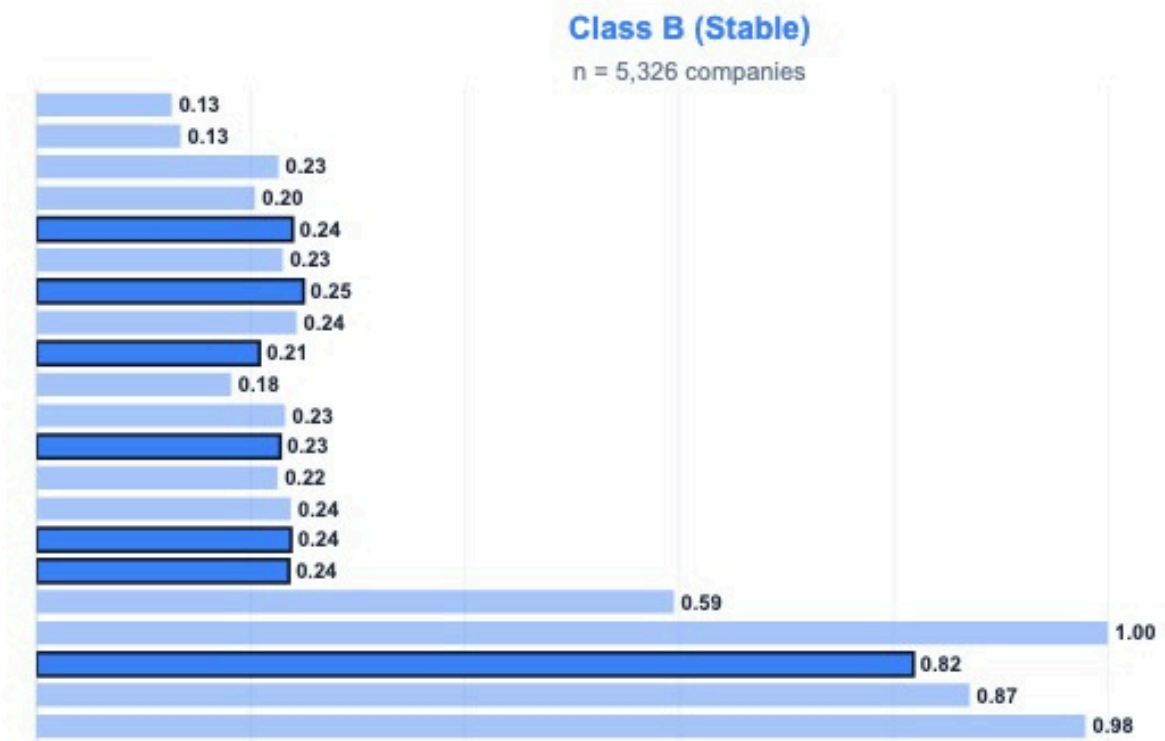
Geometric Structure

- Graph Laplacian
- Rayleigh energy λ

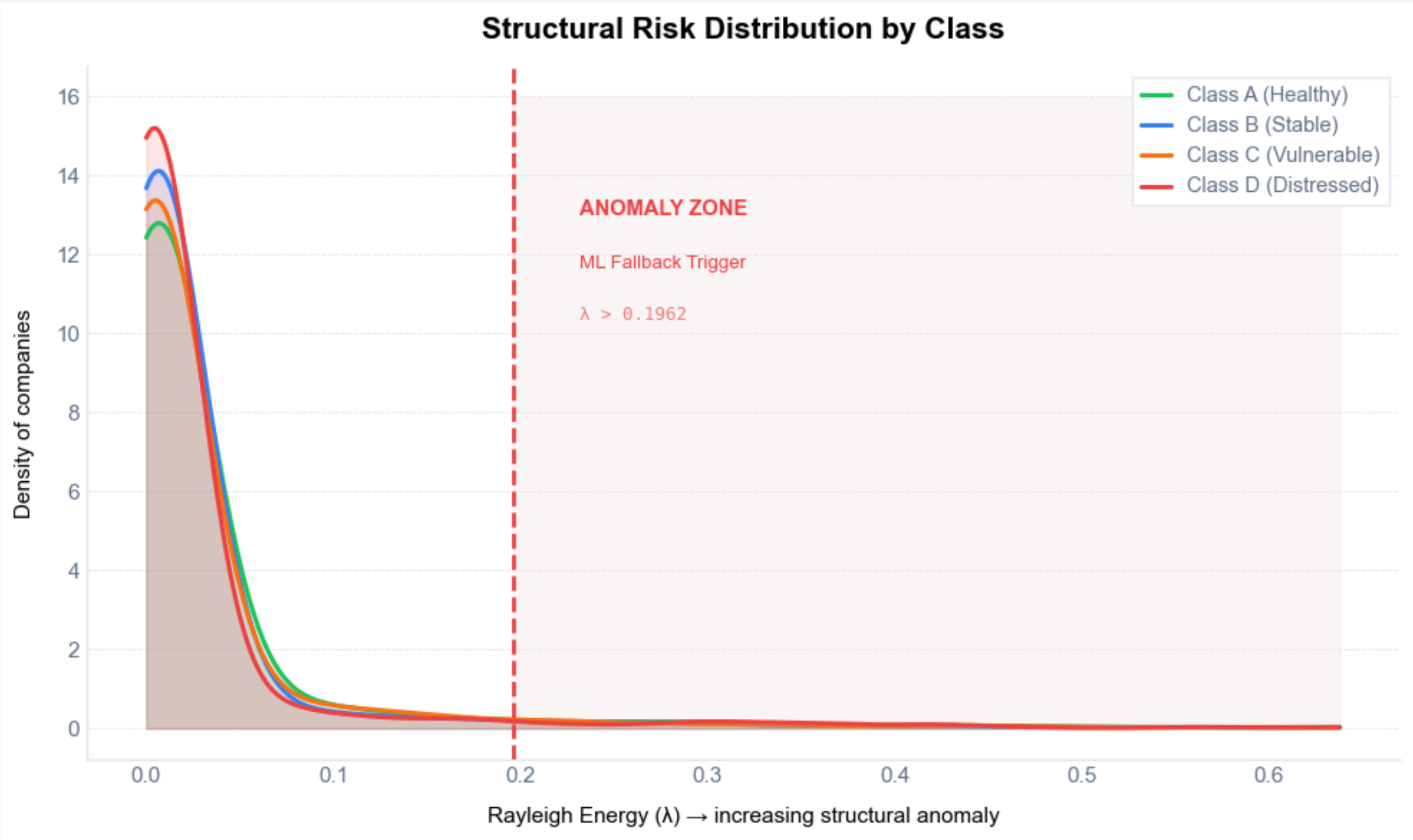
HOW THE FEATURES TALK



THE INFORMATION FLOW



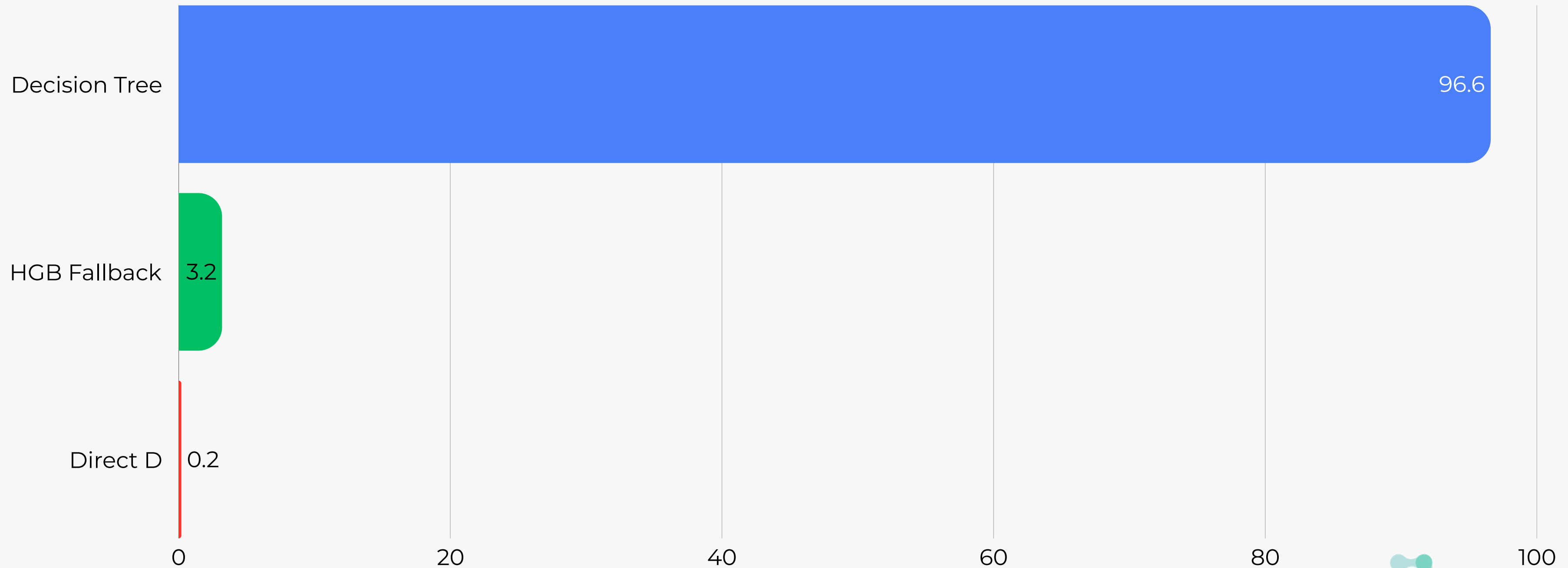
THE SPECTRAL AUDIT



Class-Level ArrowSpace Summary			
Class	Companies	Mean λ	Outliers
A Healthy	753	0.0429	31 (4.1%)
B Stable	5,326	0.0253	33 (0.6%)
C Vulnerable	2,036	0.0279	32 (6.2%)
D Distressed	756	0.0367	27 (9.2%)
n = 8,871 total		threshold λ = 0.1959	

ROUTING ARCHITECTURE

Test Set Routine
(5,811 companies)



SPECTRAL OUTLIERS

Machine Learning deployed surgically on complex anomalies (3.3%)



96.7%

Handled by Decision Tree

Fully auditable, deterministic rules

3.3%

Routed to HGB Fallback

High- λ anomalies needing ML precision

TEST PREDICTIONS

500

Class A

8.6% Healthy

3'452

Class B

59.4% Stable

1'365

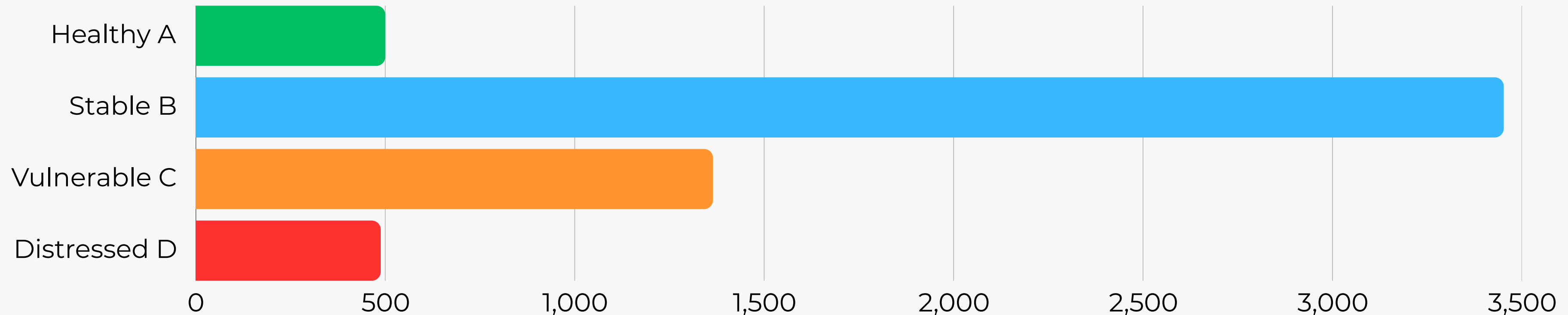
Class C

23.5% Vulnerable

488

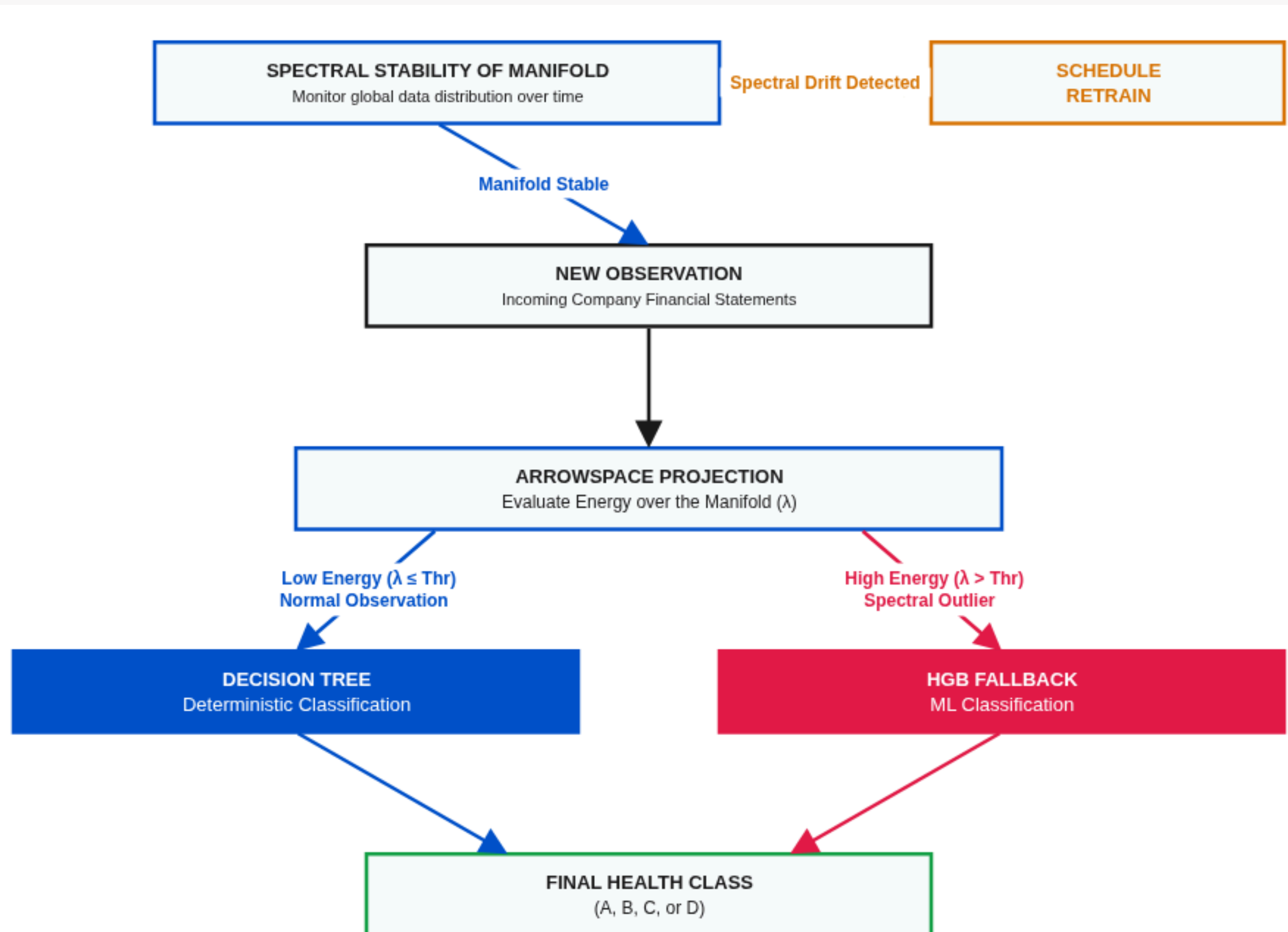
Class D

8.4% Distressed



162 companies flagged as spectral anomalies, not necessarily distressed, but structurally unusual and recommended for manual review.

THE FINAL PIPELINE



THANK YOU

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